

Optical design and synthetic analysis of the electron cyclotron emission imaging diagnostic of HL-2M tokamak

Min Jiang^{a,*}, Yilun Zhu^b, Xin Yu^a, Zhongbing Shi^a, N.C. Luhmann Jr.^b, Zhoujun Yang^c, Weichu Deng^a, Zengchen Yang^a, Yu Zhou^a, Ruihai Tong^a

^a Southwestern Institute of Physics, P.O. Box 432, Chengdu, 610041, China

^b Davis Millimeter Wave Research Center, University of California Davis, Davis, 95616, CA, USA

^c International Joint Research Laboratory of Magnetic Confinement Fusion and Plasma Physics, state Key Laboratory of Advanced Electromagnetic Engineering and Technology, School of Electrical and Electronic Engineering, Huazhong University of Science and Technology, Wuhan, 430074, China

ARTICLE INFO

Keywords:

Electron cyclotron emission imaging
Optical system
Synthetic ECEI model
HL-2M tokamak

ABSTRACT

The electron cyclotron emission imaging (ECEI) diagnostic is a powerful tool to study the MHD and turbulent transport in magnetically confined fusion plasmas. In this work, the optical system including the local oscillator (LO) coupling and radio frequency (RF) receiving optics, has been designed and analyzed for the HL-2M ECEI diagnostic. The LO optics can illuminate the antenna array and drive the mixer diode to work efficiently, with more than 36% of the beam intensity for the channels which probe the plasma edge relative to those of channels which probe the plasma core. The RF optics aims at guiding the plasma emission signal to the antenna array. To meet different physical requirements, three types of field of view have been achieved for the plasma imaging, with zoom factors of around 1, 1.5 and 2, respectively. The focal surfaces are almost flat, with a maximum off-mapping (defined as the radial distance between the beam waists of the lower-/uppermost antennas and electron cyclotron emission layer) less than 2.5 cm, which can match the electron cyclotron emission layer quite well and suppress the image distortion in the plasma edge region. The impact of possible vibrations and installation error on the focal plane has been evaluated. In addition, the predefined MHD and turbulence perturbations are well reproduced by combining the optical simulation results and synthetic ECEI modeling, which further verifies the good performance of the RF optics.

1. Introduction

In the magnetically confined fusion plasma, one of the unresolved issues is the clarification of the instabilities, such as macro-scale magnetohydrodynamics (MHD) and micro-scale turbulence. Under conditions of favorable optical thickness, the two dimensional local measurement of the electron temperature fluctuations ($\delta T_e(R, z)$, where R and z are the radial and vertical locations, respectively) can be obtained from the electron cyclotron emission imaging (ECEI) diagnostic by collecting the second harmonic X-mode radiation with high temporal and spatial resolutions [1,2]. The ECEI has been developed in many tokamaks, such as TEXTOR [3], KSTAR [4], DIII-D [5,6], ASDEX-Upgrade [7], EAST [8], HL-2A [9,10] and J-TEXT [11,12], and has been demonstrated as a powerful and state of the art tool to study the MHD activity (such as tearing modes [13,14], sawteeth [15,16], and Alfvén eigenmodes [17,18]) and recently also multi-scale MHD-turbulence interactions [19–21]. In addition, in regions of low optical thickness, such as the plasma edge, the ECE emission could be dominated by

density fluctuations, and in this case the ECEI diagnostic can be used to explore the pedestal dynamics with the help of radiation transport modeling [4,22].

The optical system, as the key component of the ECEI diagnostic, determines the spatial resolution, field of view, and the signal-to-noise ratio of the $\delta T_e(R, z)$ images. In this work, the ECEI optical system for the HL-2M tokamak [23] has been designed and assessed, including the local oscillator (LO) coupling optics and radio frequency (RF) receiving optics. In addition, the imaging performance of the optics has been further verified by the advanced numerical synthetic modeling. HL-2M is designed to have flexible and advanced divertor configurations, in order to deal with large power fluxes ($>10 \text{ MW m}^{-2}$). It is a medium-sized tokamak with an aspect ratio of 2.8, plasma current $I_p \leq 3 \text{ MA}$, toroidal field $B_{t0} \leq 3 \text{ T}$, major radius $R_0 = 178 \text{ cm}$, minor radius $a = 65 \text{ cm}$, elongation $\kappa \leq 1.8$ and triangularity $\delta \leq 0.5$.

Following the introduction section, the overall design of the ECEI diagnostic and physical requirements are described in Section 2. Optical

* Corresponding author.

E-mail address: jiangm@swip.ac.cn (M. Jiang).

<https://doi.org/10.1016/j.fusengdes.2023.113570>

Received 25 June 2022; Received in revised form 10 February 2023; Accepted 13 February 2023

Available online 20 February 2023

0920-3796/© 2023 Elsevier B.V. All rights reserved.

design and performance evaluation, including the LO coupling and RF receiving optics, are presented in Section 3. Section 4 describes the results of imaging prediction by synthetic ECEI model. Finally, a summary is given in Section 5.

2. Overall design of the ECEI diagnostic and physical requirements

ECEI diagnostic measures the electron temperature (T_e) by receiving the plasma electron cyclotron emission intensity (I_{ECE}) originating in the gyro motion of electrons in the magnetically confined fusion device. I_{ECE} is expressed in the Rayleigh-Jeans limit of Planck's law as $I_{ECE}(\omega) = \omega^2 k_B T_e / (8\pi^3 c^3)$, where $\omega = 2\pi f$ is the electron cyclotron frequency, c is the speed of light and k_B is Boltzmann constant. A frequency resolved measurement of the electron cyclotron emission intensity corresponds to a radially resolved measurement of the electron temperature.

As shown in Fig. 1, the ECEI diagnostic is always comprised of optical system, antenna array, electronics and digitizers [2]. The optical system includes RF optics and LO optics, as shown in Fig. 1(b). In front of the antenna array, a notch filter [24] is installed to suppress the stray electron cyclotron resonant heating power at 105 GHz by 60 dB and protect the antenna array, and the replaceable dichronic plate with different frequency is functioned as high pass filter to ensure that single sideband radiation above the LO frequency is supplied to the antenna array. Fig. 1(c) is the CAD drawing and photograph of the antenna array. The antenna array consists of 24 vertically aligned dual dipole antennas and mixer diodes (integrated Schottky diodes), and two linear arrays of 12 miniature elliptical substrate lenses (mini-lenses, with diameter of 25.4 mm and Gaussian beam waist radius of 11 mm) in front of them, named "odd channel array" and "even channel array", as shown in Fig. 1(c). The mini-lens array covers the height of 330 mm (25.4 mm $\times$ 13). The antenna array box has two orthogonal ports which allow LO beam (LO power collected by LO optics) and RF beam (ECE radiation collected by RF optics) to pass through, respectively. Inside the array box, both the LO beam and RF beam are separated into two paths by a 3 dB dielectric beamsplitter before collected by the odd channel and even channel mini-lenses. Antennas in this arrangement receive the reflected (transmitted) RF beam and the transmitted (reflected) LO beam and image to the same location in the plasma. The combined LO and RF signals are down-converted to intermediate frequency (IF) signal with frequency of 2–9.2 GHz by the mixer diodes in the antenna array, see the block diagram in Fig. 1(a).

In the electronics box, each IF signal is further double sideband down-converted and frequency resolved by eight different LO sources with the techniques of tunable voltage-controlled oscillators (VCO), with central frequencies at $f_{VCO} = 2.5, 3.4, 4.3, 5.2, 6.1, 7.0, 7.9$, and 8.8 GHz or $f_{VCO} = 3.32, 4.0, 4.6, 5.2, 5.8, 6.4, 7.0$ and 7.7 GHz. Such double sideband conversion will not affect the ECE measurement principle (one plasma volume is probed according to the emission layer at a given frequency), as well as the following modeling and prediction, since at a given LO frequency the probed plasma volume and radial resolution depend on the individual VCO frequencies and the bandwidth of the bandpass filter after the mixer. For example, for $f_{LO} = 105$ GHz, each IF channel measures the plasma volume of f (GHz) = 105 + $f_{VCO} \pm W_{BP}$ (remember that single sideband radiation above the LO frequency is supplied to the antenna array), where the W_{BP} is the bandwidth of the bandpass filter. Consequently, an effective RF bandwidth of $2W_{BP}$ (700 MHz) determines the spatial resolution of the ECEI measurement, which is ~ 1 cm under the typical operational condition. The IF signals are amplified, low pass filtered (50 kHz, 100 kHz, 200 kHz and 400 kHz), detected, and finally transported to the digitizers. Therefore, the two dimensional image with 24 (vertical) $\times$ 8 (radial) = 192 spatial locations can be obtained by the ECEI.

The physical requirements of HL-2M ECEI diagnostics are as follows:

(i) The routinely operational magnetic field of HL-2M is 1.6–2.2 T, so a wide LO frequency range is expected to investigate the structure both in plasma core (e.g., sawtooth crash, Alfvén eigenmode) and edge regions (e.g., edge localized modes). Fig. 2 shows the frequencies (f) of the 2nd and 3rd harmonic X-mode ECE radiation as a function of minor radii (r) of HL-2M plasmas, with the operational magnetic field of 1.6–2.2 T. In the regions of $r < -20$ cm, the 2nd harmonic radiation in the high field side (HFS) overlaps with 3rd harmonic in the low field side, which makes it difficult to interpret the received emission in that region. Therefore, the LO frequency is set at 75–135 GHz to cover $-20 \text{ cm} \leq r \leq 60 \text{ cm}$ for 1.6–2.2 T operation, as illustrated by the red box in the figure. To cover the LO frequency range, two horns are used, with one W-band, and the other F-band. Note that, the measurement range of ECE can be significantly extended towards the HFS by installing the receiving waveguide at the HFS wall, as calculated by the synthetic code, which has been reported in another paper [25].

(ii) The power onto each antenna needs to be larger than 1 mW to pump the mixer diodes (for optimum operation). The total power of the LO source is about 50 mW. Therefore, the image surface of the LO optics needs to be as uniform as possible in the vertical direction to sufficiently illuminate the upper- and lowermost channels which probe the plasma edge. Also, the LO optics needs to be adjustable to ensure similar beam height on the antenna array for the whole 75–135 GHz range.

(iii) The RF optics determines the focal location, view field and image surface, which is fixed for each plasma pulse. The focal locations are required to be in accord with the frequency resolved radial locations under different LO frequencies. According to physical requirements of measuring micro-scale turbulence or macro-scale MHD modes, the RF optics will be operated in narrow zoom or wide zoom pattern. Also, the ECE layer (cold resonance layer considered here, depends only on the magnetic field strength) is very narrow in the z direction (see Fig. 1(b)), which requires the image surface (comprised by beam waists in the plasma of vertically aligned mini-lenses) to match the ECE layer as best as possible.

3. Optical design and performance evaluation

The optical design is conducted by a diffractive optical simulation module. The relationship between the coordinates of the optical system and plasma frame is as follows: optical axis in the simulation module corresponds to the radial (R or r) direction in plasma, Y axis (along the lens height, E-plane of the electromagnetic field) corresponds to the vertical (z , close to poloidal) direction in plasma, and X axis (along the lens width, H-plane of the electromagnetic field) corresponds to the toroidal direction in plasma. Table 1 summarizes the information of the optical components including both the LO and RF optics, such as the surface type, X and Y curvature radii of two surfaces of individual lenses. The material of the lenses are high-density polyethylene with refractive index of 1.59.

Several methods for the generalized beam propagation have been used to analyze the performance of the designed optics. The Gaussian Beam Trace (GBT) option traces a "slow" Gaussian beam through an optical system, using propagation equations and calculates at each surface the beam radius, beam orientation, wavefront radius of curvature, waist position and waist radius. The Real Ray Trace (RRT) displays specific ray paths in great detail, helping us determine aperture sizes and evaluate problems with reference rays that can prevent other calculations from running. The Beam Synthesis Propagation (BSP) option is diffraction-based propagation technique and uses an algorithm that models the wave nature of light through an entire optical system. The output is the amplitude, intensity, or phase of the optical field at any surface throughout the optical system or in the far field following any surface.

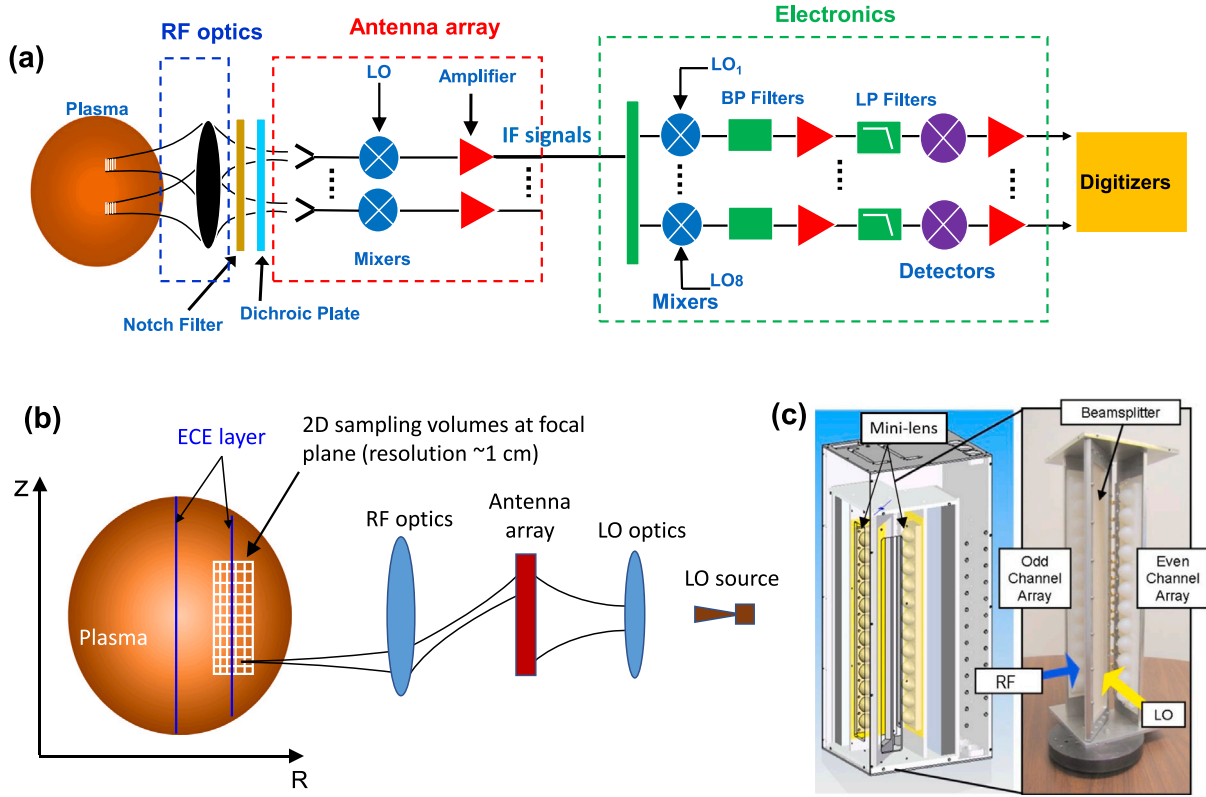


Fig. 1. (a) Block diagram of the ECEI, (b) schematic figure of ECEI optics, and (c) CAD drawing and photograph of the antenna array.

Table 1

Summary of the information of the optical components, including LO optics and RF optics. The surface type, X and Y curvature radii of two surfaces of individual lens are listed separately. σ is the conic constant of an asphere. Cyl.: cylinder, Sph.: sphere, Inf.: infinity.

	Lens name	Surface type	X curvature radii (mm)	Y curvature radii (mm)
LO optics	First lens	Cyl., Cyl.	Inf., -200	-50, Inf.
	Second lens	Cyl., Cyl.	Inf., -150	890, Inf.
RF optics	H1	Cyl., Cyl.	250, -250	Inf., Inf.
	E1	Sph., Y-toroid ($\sigma = 0.08$)	Inf., Inf.	Inf., -918.2
	FCA	Cyl., Y-toroid ($\sigma = 0.11$)	Inf., Inf.	-498.8, -1008.5
	Z1	Cyl., Cyl.	Inf., Inf.	-1598.7, -709.3
	Z2	Cyl., Sph.	Inf., Inf.	888.2, Inf.
	H2	Cyl., Cyl.	1055, -1055	Inf., Inf.

3.1. Local oscillator coupling optics

The coupling LO optics is used to guide and collimate parts of the microwave beam emitted from the horn to the vertically aligned mini-lenses in the antenna array. Three requirements should be satisfied for the LO optics: (i) beam illuminates the mini-lenses with high power (>1 mW) to drive the mixer diode in the antenna array, (ii) the beam intensity on the array is as uniform as possible to ensure similar amplitude between central and uppermost/lowermost channels, and (iii) adjustable for the whole LO frequency range (75–135 GHz) measurement. To obtain the above requirements, the LO optics is comprised of two lenses, named “first lens” and “second lens”, with surface curvature in both E-plane (Y/vertical direction) and H-plane (X/toroidal direction) as listed in Table 1.

Fig. 3 shows the layout of the LO optics. The distance between the horn and the antenna array is 1180 mm. To cover the LO frequency range, two horns are used. The W-band (75–110 GHz) horn has a Gaussian beam radius of 8.94 mm (E-plane) $\times$ 8.37 mm (H-plane),

while that of the F-band (90–140 GHz) horn is 7.26 mm (E-plane) $\times$ 6.83 mm (H-plane). The two horns are rigidly fixed on the stepping motor and can be swapped remotely between plasma pulses. The first lens is used to expand the beam in the E-plane and converge the beam in the H-plane. The second lens is used to collimate the beam in E- and H-planes to ensure the slender beam onto the antenna array.

Fig. 4(a) shows the E-plane GBT of the LO coupling optics at frequency $f \approx 105$ GHz (wavelength about 3 mm) using the W-band horn, and Fig. 4(b) shows the two dimensional distribution of the beam intensity at the antenna array predicted by BSP for $f \approx 105$ GHz. The beam radius at the antenna array is about 220 mm in E-plane and 9.2 mm in H-plane. In addition, the vertical-cut profile of beam intensity is quite flat in the center, and the relative intensity at the edges of the antenna array is more than 48% of the central intensity, which indicates that the LO beam can illuminate the whole antenna array and hence drive the mixer diodes quite well (half height of the antenna array is 165 mm).

Table 2

Calibration of the location of the first lens dependent on LO frequency. The last column is the relative intensity on the upper-/lowermost antennae compared with the maximum intensity on the central antenna.

Freq. (GHz)/ Wavelength (mm)	Distance from horn to first lens (mm)	Beam height & width on antennae (mm)	Relative intensity on antennae
75/4	50	452, 16	$\geq 36\%$
90/3.33	75	440, 16.8	$\geq 40\%$
105/2.86	107	440, 18.4	$\geq 48\%$
135/2.22	176	440, 23	$\geq 44\%$

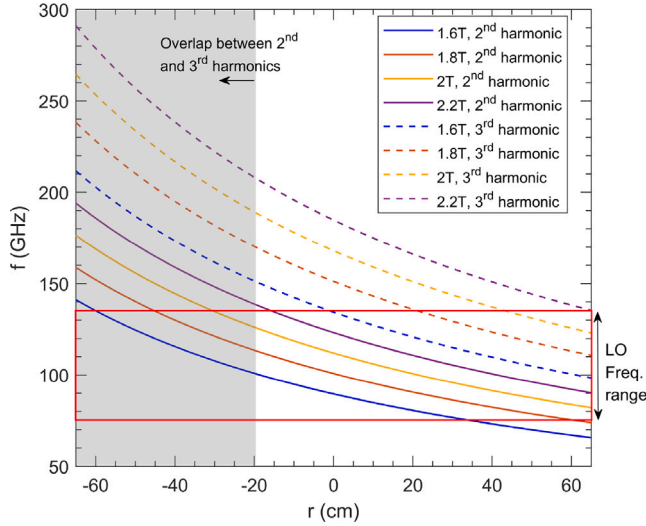


Fig. 2. Frequency f of the 2nd and 3rd harmonic X-mode ECE radiation as a function of minor radius r of HL-2M plasma at different magnetic fields. The shaded area denotes the overlap region between 2nd and 3rd harmonics.

The LO frequency of the ECEI diagnostic needs to be varied flexibly during the experimental campaign, according to the changes of the operational magnetic field and the required detection location due to different physics of interest. It is found that as the frequency increases, the beam radii in E- and H-planes on the antenna array decrease. Fortunately, this can be compensated by shifting the position of the first lens. Table 2 lists the locations of the first lens dependent on frequency to ensure that the E-plane beam radius on the antenna array is about 220 mm. The last column of Table 2 is the relative intensity on the upper-/lowermost antennae compared with the maximum intensity on the central antenna. In the frequency range 75–135 GHz, the beam intensity received by the antenna edges is larger than 36%, which is enough to drive the edge diodes (>1 mW required). The position of the first lens can be remotely controlled by the stepping motor between the plasma pulses.

3.2. Radio frequency receiving optics

According to the principle of optical path reversibility, the RF receiving optics is designed based on the Gaussian beam theory by taking the antenna array as the object and the plasma as the image surface. The Gaussian beam waist radius of the mini-lens is 11 mm. The target is to produce the one-to-one mapping relationship between the plasma volume of ECE radiation and the vertically aligned antennae. The view field of the RF optics needs to be zoomed due to different physical requirements. In addition, the image surface should be as flat as possible to match the electron cyclotron emission layer in the plasma.

Fig. 5 shows the layout of the RF optics, which includes two H-plane lenses (H1 and H2), four E-plane lenses (E1, FCA, Z1 and Z2),

and the window lens (just for sealing the flange window, without curvature). The H1 and H2 lenses are used to focus the beam in the toroidal direction. The E1 lens can shift the radial position of the image surface in the plasma. The FCA (abbreviation for “field curvature adjustment” [26]) lens is used to increase the flatness of the image surface, and hence reduce its deviation from the ECE layer, especially for those channels measuring the plasma edge. The zoom factor of the optics can be changed by moving the zoom group which consists of Z1 and Z2 lenses. More information on the lenses is summarized in Table 1.

Fig. 6(a)–(c) show the simulation results of the RF optics with the zoom factors of 1, 1.5 and 2, respectively, by placing the zoom group in different locations. The image range of the RF optics covers from the low field side edge to the high field side core, i.e., the whole region without harmonic overlapping as shown in Fig. 2. Three target focal planes located at $r = -4.3$ cm (core), 29.1 cm and 60.4 cm (pedestal) are exemplified by shifting the location of the focus lens (E1), as shown in Fig. 6(a)–(c) from the top to the bottom, respectively. In order to remotely adjust the focal plane and the field of view pulse by pulse based on the scientific requirements, the focus lens, the FCA lens and the zoom group will be installed on independent stepping motors.

The depth and shape of the image surface in the plasma (formed by the beam waists of each antenna) have been calculated by the GBT and RRT procedures. The results are shown in Fig. 7, with blue circles, red triangles and black diamonds denoting zoom factors (Z) of 1 (narrow zoom), 1.5 (intermediate zoom) and 2 (wide zoom), respectively. In the narrow zoom case, the vertical field of view is about 33–40 cm, with a vertical spatial resolution of 1.1–1.3 cm (depends on the beam waist radius in the plasma), which is favorable to investigate the micro-instability such as turbulence. In the wide zoom case, the vertical field of view is about 50–64 cm, with a vertical spatial resolution of 1.7–2.2 cm, which is more suitable to study MHD physics. One can see that the image surface is almost flat, and even at the plasma core the off-mapping (Δr , defined as the radial distance between the beam waist of the lower-/uppermost antenna and the ECE layer, as illustrated in Fig. 7) is less than 2.5 cm in the wide zoom case. It indicates that the image surface of the RF optics matches well the electron cyclotron emission layer of the 2nd harmonic domain of the X-mode wave in the plasma, which ensures good locality of the ECEI measurements, especially the spatial resolution in the plasma edge region.

The parameters including the distance from E1 lens to H1 lens and focal location r , and the image performance of the RF receiving optics including the beam radii, off-mapping of the beam waist of upper-/lowermost antenna from the ECE layer, view field and zoom factors, are shown in Table 3 in narrow, intermediate and wide zoom cases. The locations of the focal lens and zoom group can be remotely and flexibly adjusted pulse by pulse to achieve different plasma depth and field of view according to different scientific requirements. Those parameters in Table 3 can provide the reference for the laboratory calibration of the RF optics. For the instability mode with toroidal mode number of n , the wavelength of the distorted magnetic surface in the toroidal cross section is $2\pi R_0/n$, which is about 11 184 mm for $n = 1$ and 745 mm for $n = 15$. The beam radii of the image surface in the toroidal direction are 15–25 mm, which ensures a good balance between receiving enough radiation intensity and required measurement locality of instability modes. The three dimensional (radial, vertical, toroidal) structure will

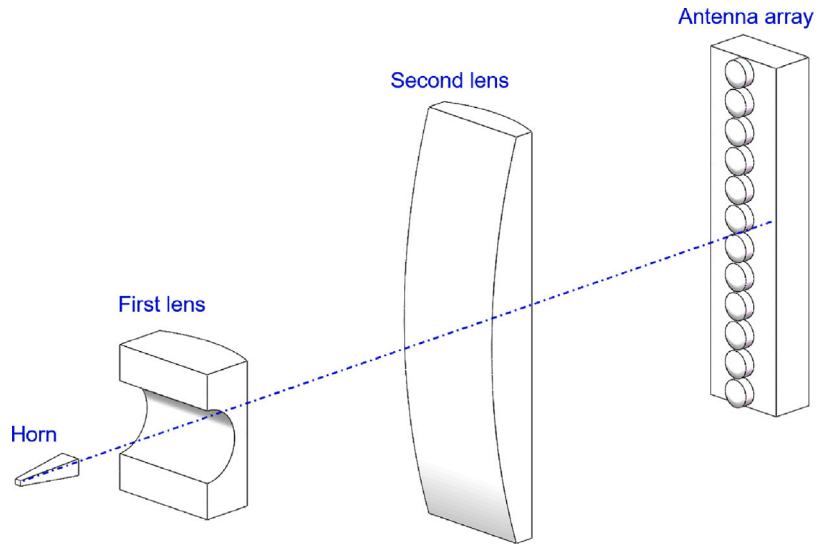


Fig. 3. Layout of the LO optics, which includes the first lens and second lens.

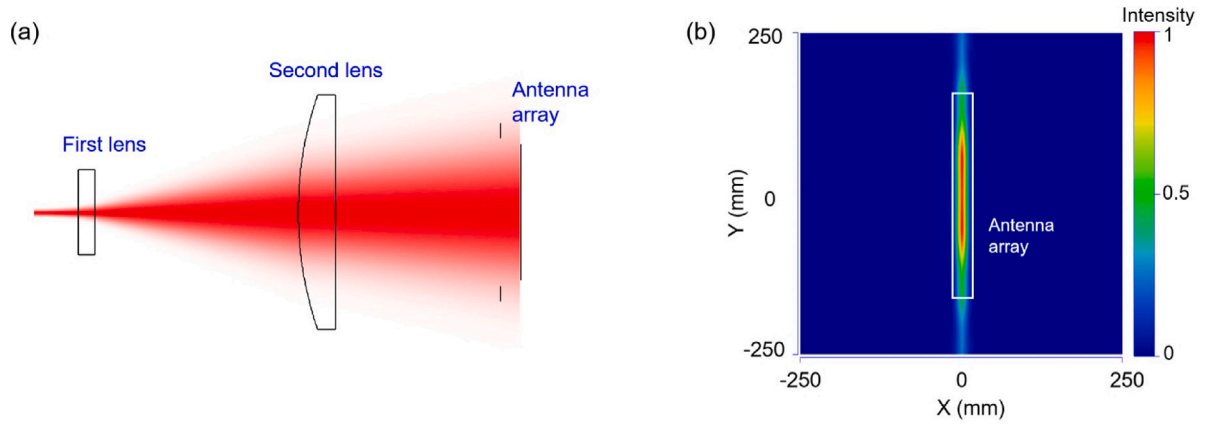


Fig. 4. (a) E-plane Gaussian Beam Trace of the LO coupling optics for the W-band horn at frequency $f \approx 105$ GHz. (b) Two dimensional distribution of the beam intensity at the antenna array by Beam Synthesis Propagation for $f \approx 105$ GHz.

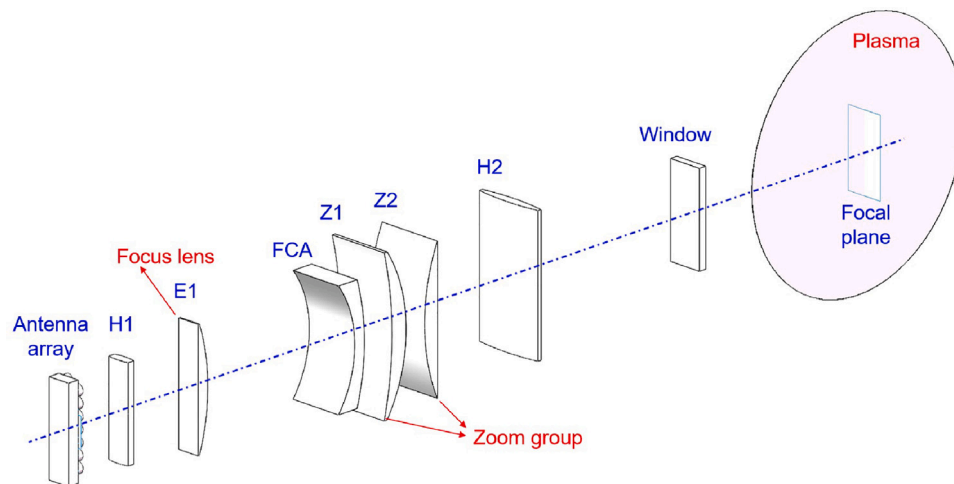


Fig. 5. Layout of the RF receiving optics, which includes two H-plane lenses (H1 and H2), four E-plane lenses (E1, FCA, Z1 and Z2) and the window lens.

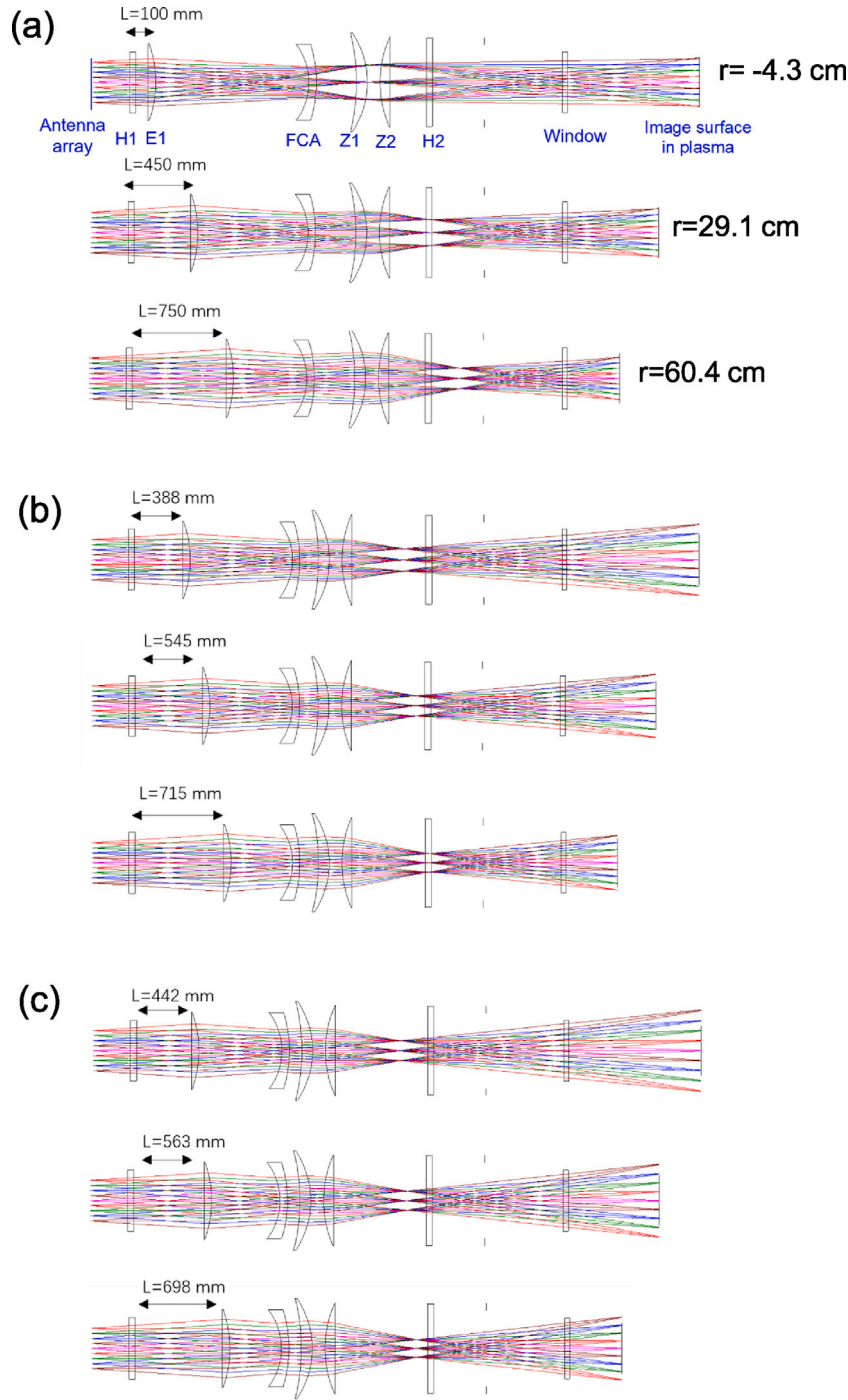


Fig. 6. (a)–(c) Simulation of the RF receiving optics with zoom factor about 1, 1.5 and 2, respectively. The image surfaces are located at $r = -4.3$ cm, 29.1 cm and 60.4 cm from top to bottom.

be detected by installing ECEI in two toroidally separated windows of the fusion device.

In reality, the vibration of the optical platform or the error of installation can bring some misalignment of the image surface with the ECE layer. Table 4 illustrates the impact of displacement and tilt of lenses on the imaging performance under the intermediate zoom case ($Z \approx 1.5$). The radial (dr) and vertical (dz) movement of the center of the image surface, and the image surface tilt which is defined as the radial distance (δr) between the images of the uppermost and lowermost antennae are concerned, when all the lenses are vertically moved upwards (“+”) and downwards (“−”) by 10 mm, or E1, FCA,

Zoom group lenses are tilted individually along the direction of the lens width by $\pm 5^\circ$ (“±” denote clockwise and anticlockwise directions, respectively). Here, two focal locations, i.e., plasma center and pedestal region, are considered. The results indicate that the image surface is significantly tilted by almost all the aforementioned changes, particularly by the FCA and Zoom group tilting (see the third and sixth rows of Table 4), which will bring serious trouble in the data interpretation. The vertical displacements by ± 10 mm (e.g., could be induced by vibration during the plasma discharge) mainly cause the focal plane to shift vertically and radially by 2–3 cm. In addition, the center of the image surface is very sensitive to the tilt of the Zoom group (see larger

Table 3

Parameters including the distance from E1 lens to H1 lens and focal location in terms of minor radius r , and the image performance including the beam radii, off-mapping Δr of the beam waist of upper-/lowermost antenna from the ECE layer, view field and zoom factors of the RF receiving optics in narrow, intermediate and wide zoom cases. Unit: mm.

Zoom pattern	Distance from E1 to H1	Focal location r	Beam radii (vertical, toroidal)	Δr	View field	Zoom factor
Narrow	100	-43	13, 25	18.8	390	1.2
	450	291	13, 15	10	382	1.18
	750	604	11.4, 24.3	24.8	335	1
Intermediate	388	-43	19, 24	9	565	1.75
	545	291	17, 15	6	507	1.57
	715	604	15, 24	3.4	440	1.37
Wide	442	-43	22.6, 25	25	640	2
	563	291	20, 15.5	9	568	1.76
	698	604	17.2, 24	0.3	492	1.53

Table 4

Impact of the displacement and tilt of the lenses on the image performance of RF receiving optics under the intermediate zoom case ($Z \approx 1.5$). The dr and dz are the radial and vertical displacements of the center of the image surface, respectively, with “+” and “-” denoting outward (upward) and inward (downward) directions. Image tilt δr is defined as the radial distance between the images of the uppermost and lowermost antennae. Lens tilt angle is defined as the rotation along the direction of the lens width, with “ $\pm$ ” denoting clockwise and anticlockwise directions, respectively.

Plasma location	Parameters of image surface	Vertically move by ± 10 mm	E1 tilts by $\pm 5^\circ$	FCA tilts by $\pm 5^\circ$	Z1 and Z2 tilt by $\pm 5^\circ$
Core	dr (mm)	-24, -24	-9, -9	27, 27	-82, -82
	dz (mm)	27, -27	-3, 3	-6, 6	-6, 6
	tilt δr (mm)	43, -43	48, -48	237, -237	-330, 330
Edge	dr (mm)	-11, -11	-14, -14	16, 16	-51, -51
	dz (mm)	23, -23	-2, 2	-4, 4	-3, 3
	tilt δr (mm)	2, -2	-26, 26	230, -230	-250, 250

4. Prediction by synthetic ECEI model

Recently, the synthetic ECEI model has been developed to better interpret experimental data, especially in the cases that the ECE radiation deviates from the blackbody spectrum in the plasma edge or the emission frequency is close to the local cutoff frequency in the high density plasma core [27,28]. It applies a reciprocity theorem for the calculation of the received ECE power, and self-consistently includes refraction, diffraction, emission and absorption effects. It relates the original problem with its reciprocal problem, where a wave is launched from the antenna, propagates into the reciprocal plasma, and is absorbed by the electron cyclotron resonance. It is shown that the received time averaged power $P_s(\omega)$ in the original problem can be expressed in terms of a source current correlation tensor $\hat{K}$ which describes the local emission property of the original plasma and the reciprocal wave field $\vec{E}^{(+)}$, i.e., $P_s(\omega) = \frac{1}{16} \int_{-\infty}^{\infty} \vec{E}^{(+)}(\omega, \vec{r}) \hat{K}(\omega, \vec{r}, \vec{r}') \vec{E}^{(+)*}(\omega, \vec{r}') d\vec{r} d\vec{r}'$.

With the combination of sophisticated diffractive optical simulations and efficient wave propagation calculation by the synthetic ECEI model, the imaging results are examined to quantify and optimize ECEI performance. The input plasma equilibrium profiles include magnetic field, electron density and temperature, as shown in Fig. 8. Here, the toroidal magnetic field is $B_{t0} = 1.8$ T, central electron density is $n_{e0} = 3 \times 10^{19} \text{ m}^{-3}$, and central electron temperature is $T_{e0} = 4$ keV. The focal location, beam waist and the field of view of RF optics in the vacuum region (without taking beam-plasma interaction into account) as shown in Section 3.2, are put into the synthetic calculation. In this calculation, considering that the line of sight perpendicular to the magnetic field line, there is no significant parallel contribution of the cyclotron emission, and therefore the Doppler broadening is neglected. When radiation transport effects are taken into account the emission position changes, thus corresponding to a warm resonance position. It is solved for the 24 vertical line of sights and 8 radial frequencies until it reaches back to the antennae. The 2D images are obtained by mapping the radiation temperature onto positions of the 24×8 warm resonances. In Fig. 8(c), a predefined eigenmode structure with 10% relative temperature perturbation ($\delta T_e / T_e = 10\%$) near R

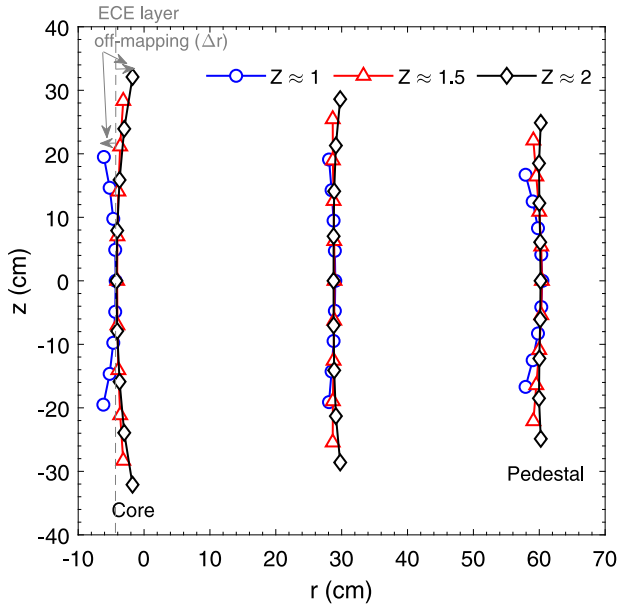


Fig. 7. Distributions of the focal points at different plasma locations under different vertical field of view. The blue circles, red triangles and black diamonds denote zoom factors (Z) of 1 (narrow zoom), 1.5 (intermediate zoom) and 2 (wide zoom), respectively.

dr), but not sensitive to the tilt of E1 and FCA lenses. Therefore, the large vertical vibration (>1 cm) and installation error (especially for the FCA and Zoom group) should be carefully addressed to ensure the performance of the RF optics.

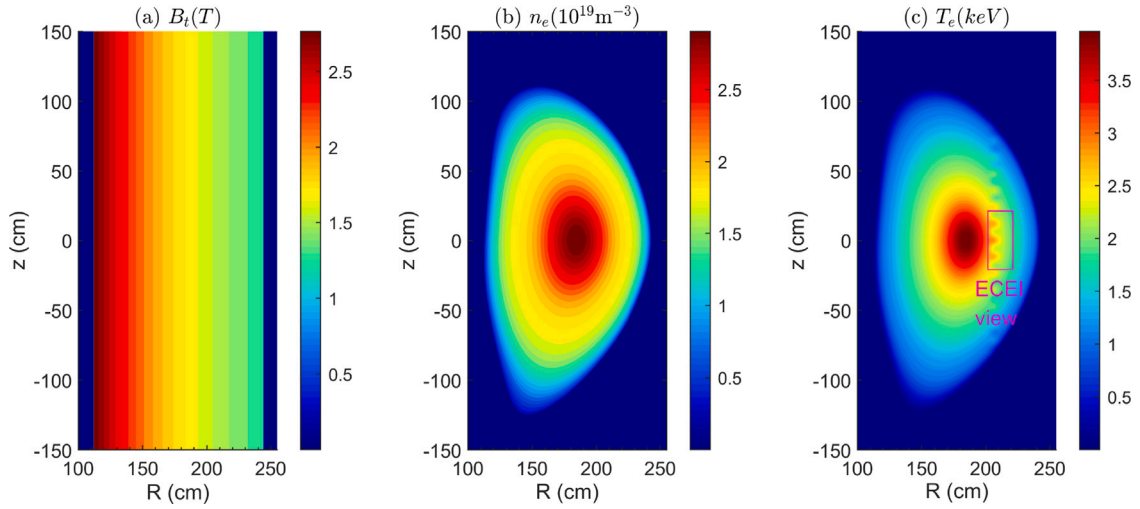


Fig. 8. The input profiles for synthetic ECEI diagnostic analysis: (a) magnetic field B_t with $B_{t0} = 1.8$ T, (b) electron density n_e with $n_{e0} = 3 \times 10^{19} \text{ m}^{-3}$ and (c) the electron temperature T_e with $T_{e0} = 4$ keV and predefined mode structure with $\delta T_e/T_e = 10\%$ added near $R = 207$ cm. The ECEI view region is marked in (c).

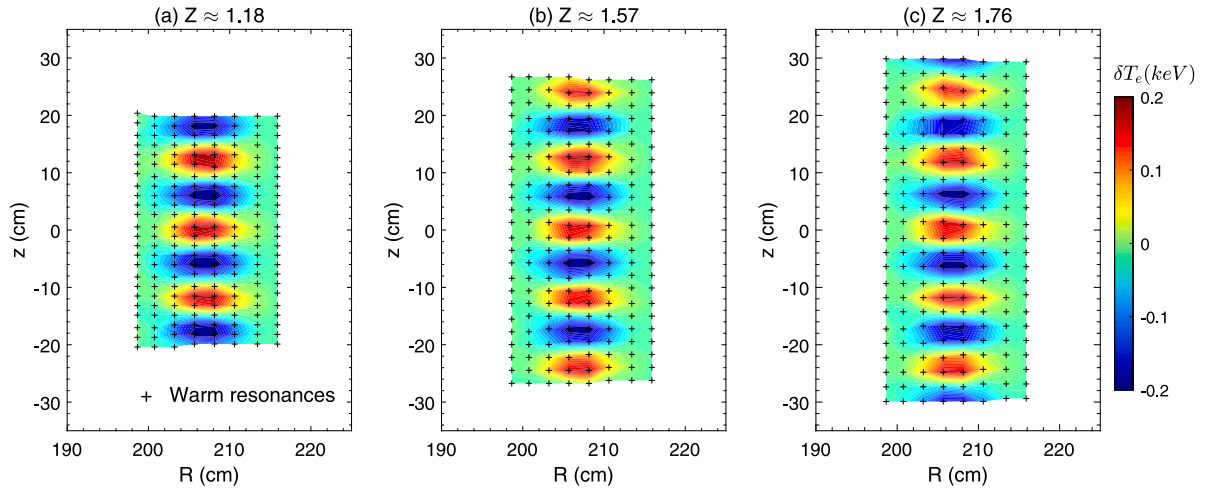


Fig. 9. Synthetic ECEI results of the MHD mode structure with the predefined perturbation of $\delta T_e/T_e = 10\%$ as shown in Fig. 8(c). (a) The narrow zoom case, (b) intermediate zoom case and (c) wide zoom case. The colorbar denotes the perturbed electron temperature δT_e , where $\delta T_e = T_{e,total} - T_{e,eq}$, with $T_{e,total}$ the output of the synthetic ECEI and $T_{e,eq}$ the equilibrium temperature shown in Fig. 8(c). The "+" symbols illustrate the positions of the 24×8 ECEI warm resonances. Simulation parameters: toroidal magnetic field $B_{t0} = 1.8$ T, central electron density $n_{e0} = 3 \times 10^{19} \text{ m}^{-3}$, and central electron temperature $T_{e0} = 4$ keV.

$= 207$ cm is added on the equilibrium T_e , and the ECEI view region is marked. The predefined temperature fluctuation is defined as $A \cdot \sin(kz - 2\pi f_0 t) \cdot e^{-(R-R_0)^2/dR^2}$, where t is the time, R and z are radial and vertical coordinates, A , R_0 , dR , k and f_0 are the relative amplitude, central location, radial characteristic length, poloidal wavenumber and rotation frequency of the perturbation structure, respectively.

Fig. 9 illustrates the mode structure calculated by the synthetic ECEI with RF optics in Section 3.2 (the predefined structure shown in Fig. 8(c)) under different zoom conditions. The colorbar denotes the perturbed electron temperature δT_e , where $\delta T_e = T_{e,total} - T_{e,eq}$, with $T_{e,total}$ the output of the synthetic ECEI and $T_{e,eq}$ the equilibrium temperature. The "+" symbols denote the positions of 24×8 ECEI warm resonances. The folding of ECE spectra with the optics is due to the refraction effect of the beam. This is obvious for the channels at the top and the bottom of the ECEI arrays (see Figs. 9(a)–(c) and 10(b)–(d)), since they encounter the strong gradients along the vertical direction under the largest angle. By comparing Figs. 8(c) and 9, it is found that the ECEI system with the aforementioned optics can well image the predefined structure with small poloidal wavenumber (compared with

the inverse ion gyro-radius). And the image quality is similar for the narrow zoom, intermediate zoom, and wide zoom cases.

In addition, the image performance of the small-scale T_e fluctuations (turbulence) is also investigated by the synthetic ECEI, as shown in Fig. 10. Fig. 10(a) is the predefined T_e fluctuations with relative perturbation amplitude of 2%, and Fig. 10(b)–(d) are the results of the synthetic ECEI diagnostic with the zoom factors of about 1.18, 1.57 and 1.76, respectively. It is found that for the narrow ($Z \approx 1.18$) and intermediate zoom case ($Z \approx 1.56$), the turbulent structure can be well diagnosed, but for the wide zoom case ($Z \approx 1.76$), the structure cannot be well resolved by the ECEI due to the spatial resolution limit as mentioned previously in Section 3.2. Note that the noise level of the turbulence measurement can be significantly reduced by using two-point correlation method for adjacent poloidal channels.

5. Summary

In this work, the optical system of the ECEI diagnostic, including the LO coupling optics and RF receiving optics, has been designed for

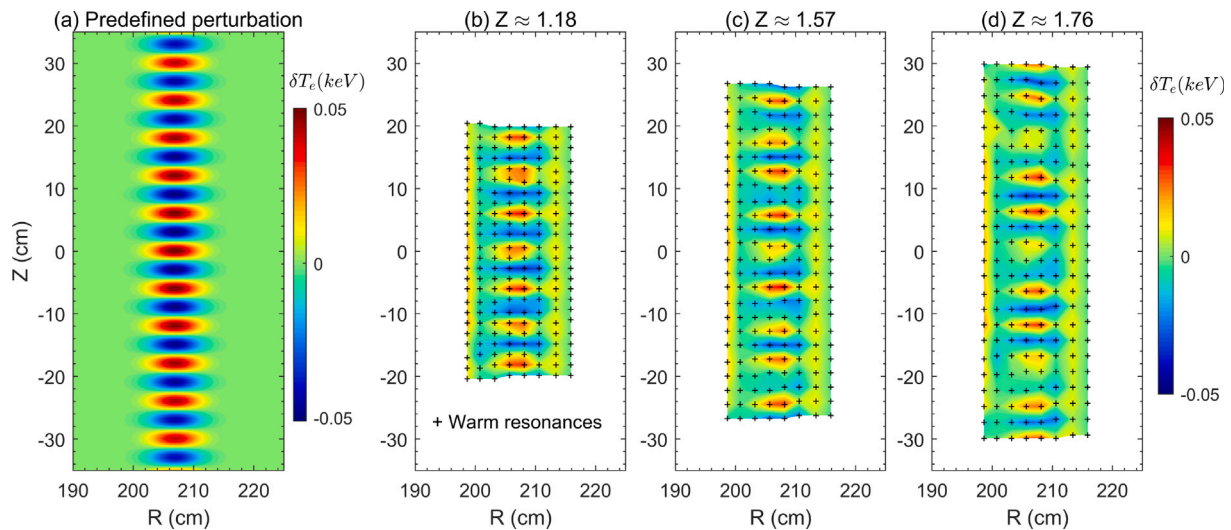


Fig. 10. Synthetic ECEI results of the turbulence structure with the predefined perturbation. (a) The predefined perturbation structure with $\delta T_e/T_e = 2\%$, and (b)–(d) results of the synthetic ECEI in the narrow, intermediate and wide zoom cases, respectively. Simulation parameters: toroidal magnetic field $B_0 = 1.8$ T, central electron density $n_{e0} = 3 \times 10^{19} \text{ m}^{-3}$, and central electron temperature $T_{e0} = 4$ keV.

the HL-2M tokamak. The LO optics well illuminates the antenna array and drives the mixer diodes, with the beam intensity onto the upper-/lowermost antennas which probe plasma edge being larger than 36% relative to that of central antennae which probe plasma core. The focal region of the RF optics can cover the whole region of interest (without harmonic overlapping), and can also work in the narrow, intermediate and wide zoom patterns for different physical requirements, such as macro-scale MHD or micro-scale turbulence investigations. Almost all the image surfaces are quite flat with a maximum off-mapping about 2.5 cm in the wide zoom case, which match well with the electron cyclotron emission layers in the plasma, and hence eventually lead to good locality and spatial resolution of the ECEI measurement. The impact of the possible vibration and installation error on the image surface has been evaluated, where the result suggests that the installation of the FCA and Zoom group should be carefully addressed. Besides, the predictive ECE images have been obtained by combining the optical simulation results and synthetic ECEI modeling for the MHD and turbulence structures, which further verifies the good poloidal spatial or wavenumber resolution of the RF receiving optics.

CRediT authorship contribution statement

Min Jiang: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition. **Yilun Zhu:** Software, Validation, Resources, Writing – review & editing. **Xin Yu:** Software, Validation, Resources, Data curation, Formal analysis. **Zhongbing Shi:** Supervision, Funding acquisition. **N.C. Luhmann Jr.:** Software, Funding acquisition. **Zhoujun Yang:** Writing – review & editing, Conceptualization. **Weichu Deng:** Visualization, Data curation, Formal analysis. **Zengchen Yang:** Methodology. **Yu Zhou:** Conceptualization, Software. **Ruihai Tong:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work is partly supported by the National MCF energy R&D Program of China (Grant Nos. 2022YFE03060000 and 2019YFE03020000), partly by the National Natural Science Foundation of China, China (Grant Nos. 12075079, 11820101004 and 12005144), partly by Sichuan Science and Technology Program (Nos. 2021JDJQ0029 and 2020JDJQ0070), partly by Youth Talent Program by China National Nuclear Corporation, partly by the Young Elite Scientists Sponsorship Program by CAST, China (Grant No. 2018QNR0001), and partly by the U.S. Department of Energy, United States under DE-FG02-99ER54531.

References

- [1] I.H. Hutchinson, *Principles of Plasma Diagnostics*, Cambridge University Press, New York, 1987.
- [2] N.C. Luhmann, et al., *Fusion Sci. Technol.* 53 (2008) 335.
- [3] H.K. Park, et al., *Rev. Sci. Instrum.* 75 (2004) 3787.
- [4] G.S. Yun, et al., *Rev. Sci. Instrum.* 81 (2010) 10D930.
- [5] B. Tobias, et al., *Rev. Sci. Instrum.* 81 (2010) 10D928.
- [6] Y. Zhu, et al., *Rev. Sci. Instrum.* 91 (2020) 093504.
- [7] I.G.J. Classen, et al., *Rev. Sci. Instrum.* 85 (2014) 11D833.
- [8] Y.L. Zhu, et al., *Rev. Sci. Instrum.* 87 (2016) 11D901.
- [9] M. Jiang, et al., *Rev. Sci. Instrum.* 84 (2013) 113501.
- [10] M. Jiang, et al., *Rev. Sci. Instrum.* 86 (2015) 076107.
- [11] Y. Zhu, et al., *Rev. Sci. Instrum.* 85 (2014) 11D854.
- [12] X.M. Pan, et al., *Rev. Sci. Instrum.* 87 (2016) 11E106.
- [13] I.G.J. Classen, et al., *Phys. Rev. Lett.* 98 (2007) 035001.
- [14] M.J. Choi, et al., *Nucl. Fusion* 54 (2014) 083010.
- [15] H.K. Park, et al., *Phys. Rev. Lett.* 96 (2006) 195003.
- [16] X.Y. Xu, et al., *Plasma Phys. Control. Fusion* 52 (2010) 015008.
- [17] B. Tobias, et al., *Phys. Rev. Lett.* 106 (2011) 075003.
- [18] I.G.J. Classen, et al., *Plasma Phys. Control. Fusion* 53 (2011) 124018.
- [19] M.J. Choi, et al., *Nucl. Fusion* 57 (2017) 126058.
- [20] M. Jiang, et al., *Nucl. Fusion* 58 (2018) 026002.
- [21] M. Jiang, et al., *Nucl. Fusion* 59 (2019) 066019.
- [22] B. Vanovac, et al., *Plasma Phys. Control. Fusion* 60 (2018) 045002.
- [23] X.R. Duan, et al., *Nucl. Fusion* 62 (2022) 042020.
- [24] Z. Shen, et al., Frequency selective surface notch filter for use in a millimeter wave imaging system, in: *Proc. IEEE Antennas and Propagation Society Int. Symp.*, 2006.
- [25] X. Yu, et al., *Rev. Sci. Instrum.* 93 (2022) 083518.
- [26] Y. Zhu, et al., *J. Instrum.* 11 (2016) P01004.
- [27] L. Shi, et al., *Rev. Sci. Instrum.* 87 (2016) 11D303.
- [28] M. Cheng, et al., *Rev. Sci. Instrum.* 89 (2018) 10H117.